

<https://github.com/joansebastiasalas/BDD-JoanSebastianSalasSuau/tree/main/BDD-Tasks>

Exercise 1

• Insert a new record in the customers table using the graphical tool offered by MySQL Workbench. The data must be the following:

Field Value

customerNumber 500

customerName Stellar Tech Ltd.

contactLastName Rivers

contactFirstName Julia

phone +34 600123456

addressLine1 99 Innovation Rd.

addressLine2 NULL

city Barcelona

state NULL

postalCode 08018

country Spain

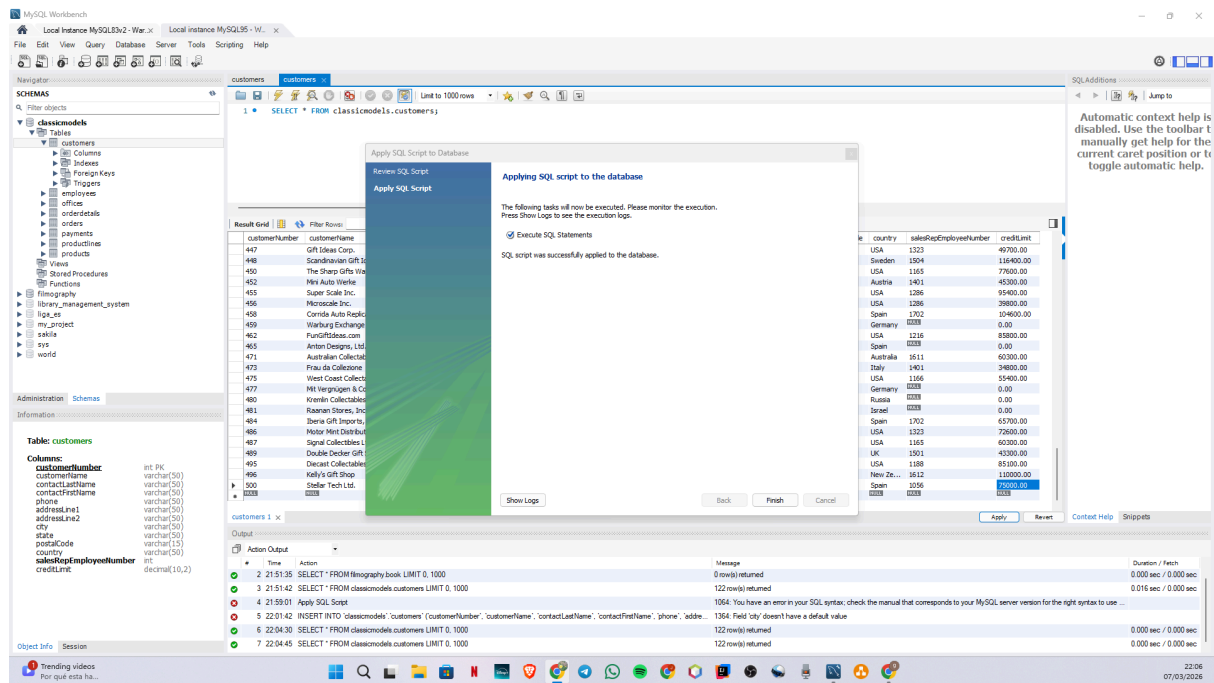
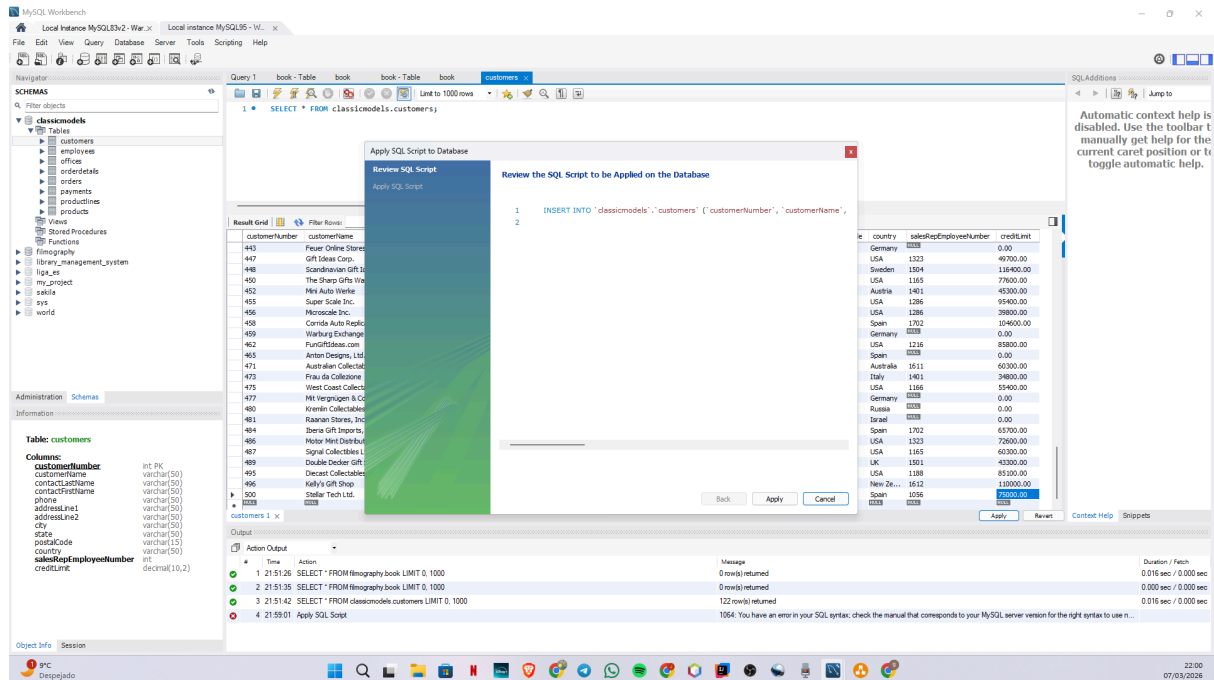
salesRepEmployeeNumber 1056

creditLimit 75000.00

- Here we have already added the necessary data provided by the exercise.

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the 'classicmodels' database structure, including tables like customers, employees, orders, and products. The 'customers' table is selected. The main window shows the 'customers' table with columns: customerNumber, customerName, contactLastName, contactFirstName, phone, addressLine1, addressLine2, city, state, postalCode, country, salesRepEmployeeNumber, and creditLimit. A new record (customerNumber 500) is highlighted in blue. The record details are: customerName: Stellar Tech Ltd., contactLastName: Rivers, contactFirstName: Julia, phone: +34 600123456, addressLine1: 99 Innovation Rd., addressLine2: NULL, city: Barcelona, state: NULL, postalCode: 08018, country: Spain, salesRepEmployeeNumber: 1056, and creditLimit: 75000.00. The bottom pane shows the 'Output' window with the SQL query: 'SELECT * FROM classicmodels.customers;' and the result: '122 row(s) returned'.

- Having filled in the previous row, we click “Apply” and “Finish”



• Insert a new record into the employees table using the graphical tool. The data must be:

Field Value

employeeNumber 2001

lastName Ortega

firstName Marc

extension x201

email marc.o@classicmodels.com

officeCode 1

reportsTo 1002

jobTitle Sales Rep

- As in the customers exercise, we added the line.

The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' pane shows the 'classicmodels' database with various tables. The 'employees' table is selected. The main window displays the 'employees' table with columns: employeeNumber, lastName, firstName, extension, email, officeCode, reportsTo, and jobTitle. A query is executed in the 'SQL' tab: `SELECT * FROM classicmodels.employees;`. The 'Output' pane shows the query execution log with the following entries:

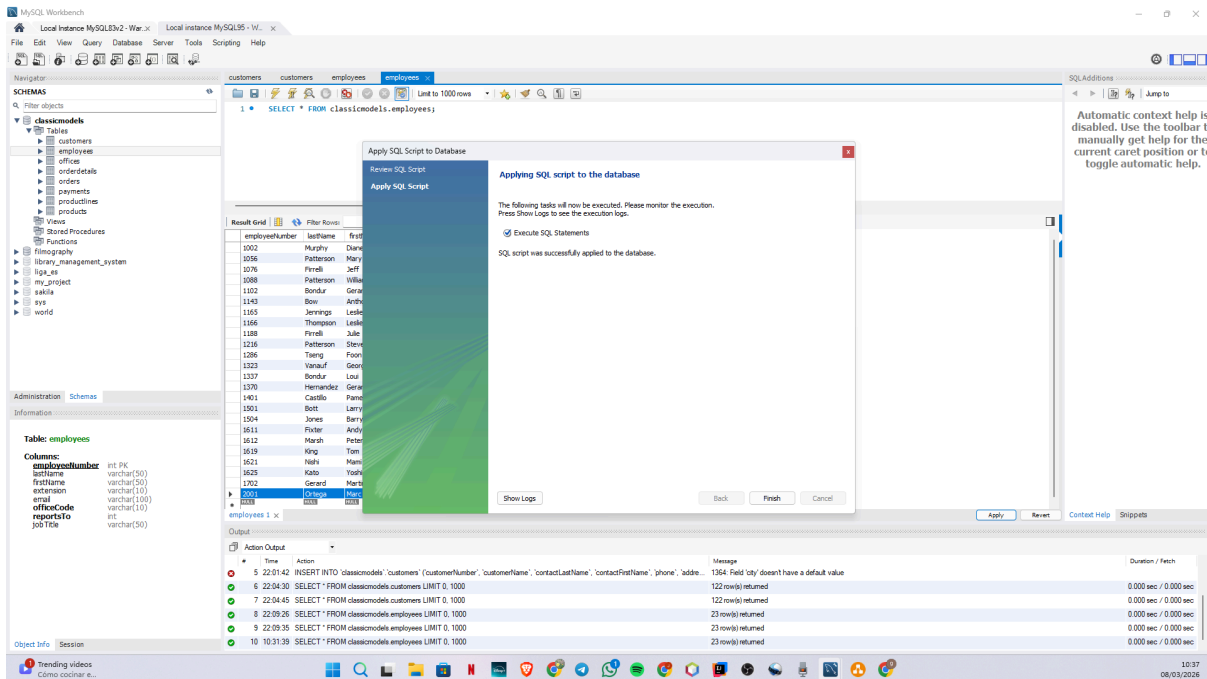
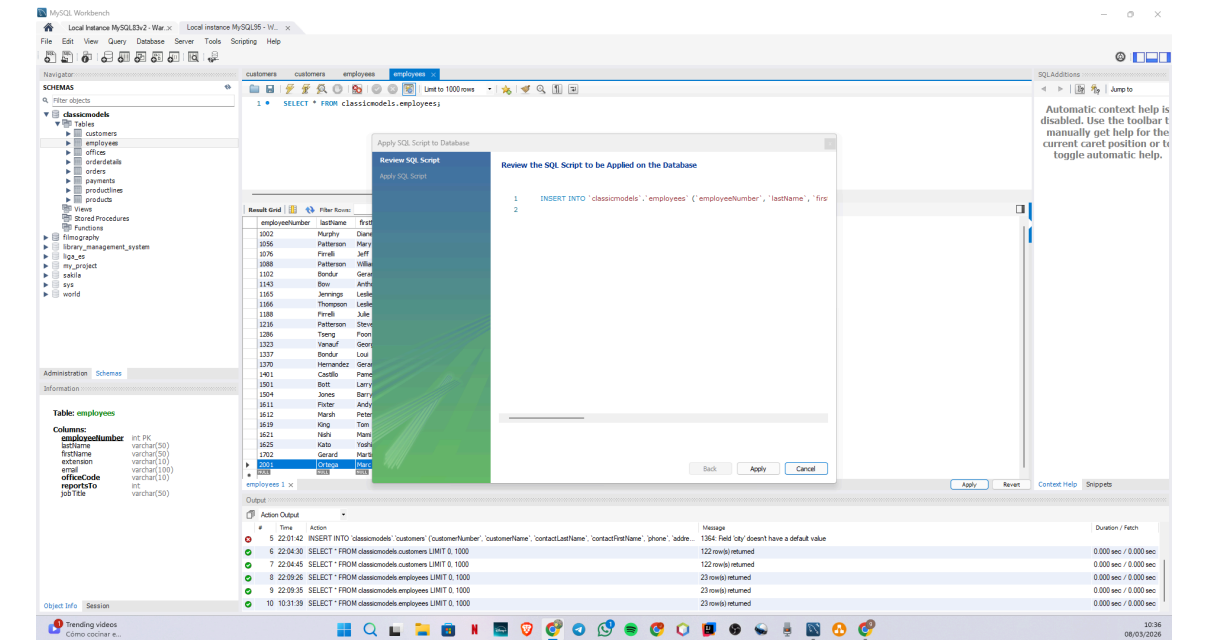
#	Time	Action	Message	Duration / Fetch
5	22:01:42	INSERT INTO 'classicmodels' 'customers' ('customerNumber', 'customerName', 'contactLastName', 'contactFirstName', 'phone', 'address...'	1364 'Field 'city' doesn't have a default value	
6	22:04:30	SELECT * FROM classicmodels.customers LIMIT 0, 1000	122 rows returned	0.000 sec / 0.000 sec
7	22:04:45	SELECT * FROM classicmodels.customers LIMIT 0, 1000	122 rows returned	0.000 sec / 0.000 sec
8	22:09:26	SELECT * FROM classicmodels.employees LIMIT 0, 1000	23 rows returned	0.000 sec / 0.000 sec
9	22:09:30	SELECT * FROM classicmodels.employees LIMIT 0, 1000	23 rows returned	0.000 sec / 0.000 sec
10	10:31:39	SELECT * FROM classicmodels.employees LIMIT 0, 1000	23 rows returned	0.000 sec / 0.000 sec

- Having already filled in the line with the information provided by the exercise.

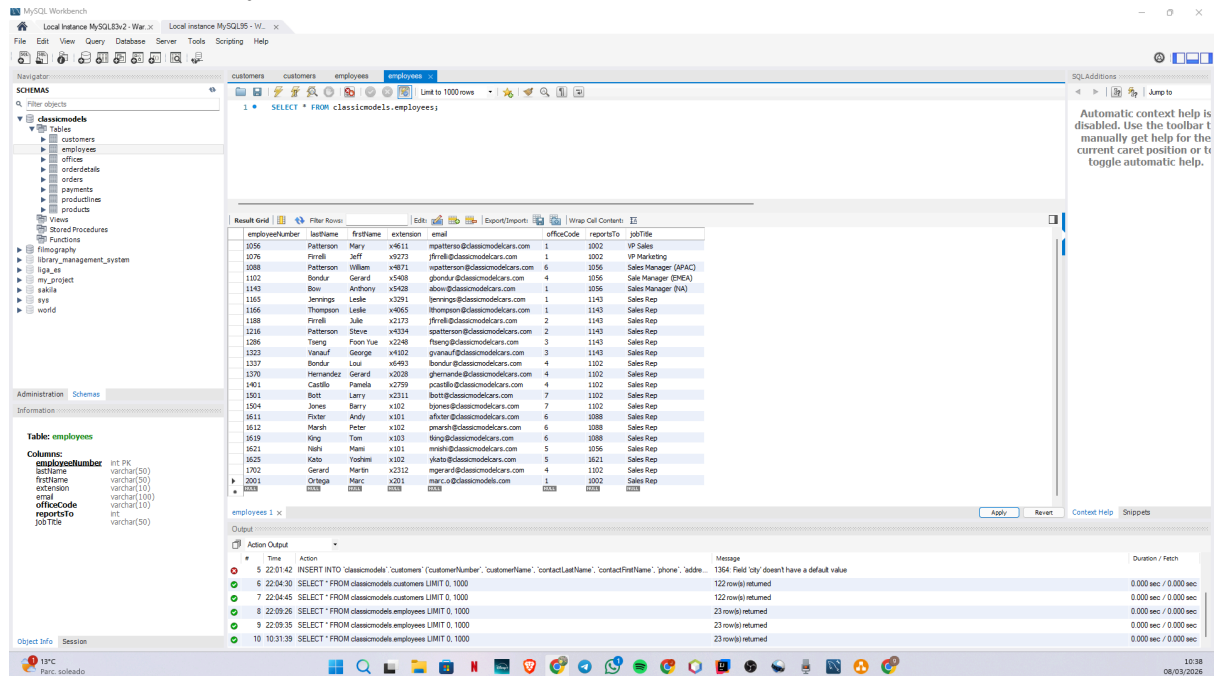
The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' pane shows the 'classicmodels' database with various tables. The 'employees' table is selected. The main window displays the 'employees' table with columns: employeeNumber, lastName, firstName, extension, email, officeCode, reportsTo, and jobTitle. A query is executed in the 'SQL' tab: `SELECT * FROM classicmodels.employees;`. The 'Output' pane shows the query execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
5	22:01:42	INSERT INTO 'classicmodels' 'customers' ('customerNumber', 'customerName', 'contactLastName', 'contactFirstName', 'phone', 'address...'	1364 'Field 'city' doesn't have a default value	
6	22:04:30	SELECT * FROM classicmodels.customers LIMIT 0, 1000	122 rows returned	0.000 sec / 0.000 sec
7	22:04:45	SELECT * FROM classicmodels.customers LIMIT 0, 1000	122 rows returned	0.000 sec / 0.000 sec
8	22:09:26	SELECT * FROM classicmodels.employees LIMIT 0, 1000	23 rows returned	0.000 sec / 0.000 sec
9	22:09:30	SELECT * FROM classicmodels.employees LIMIT 0, 1000	23 rows returned	0.000 sec / 0.000 sec
10	10:31:39	SELECT * FROM classicmodels.employees LIMIT 0, 1000	23 rows returned	0.000 sec / 0.000 sec

- Now we click on “Apply” > “Finish”.



- Line already added.



Exercise 2

Insert the following payments into the payments table using SQL statements:

customerNumber checkNumber paymentDate amount

124 H123 2024-02-06 845.00

151 H124 2024-02-07 70.00

112 H125 2024-02-05 1024.00

Copy the SQL statements you used in your submission. Only include the data shown.

- Statements

INSERT INTO payments (customerNumber, checkNumber, paymentDate, amount)

VALUES

(124, 'H123', '2024-02-06', 845.00),

(151, 'H124', '2024-02-07', 70.00),

(112, 'H125', '2024-02-05', 1024.00);

payments

```

1 INSERT INTO payments (customerNumber, checkNumber, paymentDate, amount)
2 VALUES
3 (124, 'H123', '2024-02-06', 845.00),
4 (151, 'H124', '2024-02-07', 70.00),
5 (112, 'H125', '2024-02-05', 1024.00);

```

Result Grid

	customerNumber	checkNumber	paymentDate	amount
146	FP549817	2004-03-18	40978.53	
146	FU793410	2004-01-16	49614.72	
146	LJ160635	2003-12-10	39712.10	
148	BI507030	2003-04-22	44380.15	
148	DD635282	2004-08-11	2611.84	
148	KM172879	2003-12-26	105743.00	
148	ME497970	2005-03-27	3516.04	
151	BF686658	2003-12-22	58793.53	
151	GB852215	2004-07-26	20314.44	
151	IP568906	2003-06-18	58841.35	
151	KI884577	2004-12-14	39964.63	
157	HI618861	2004-11-19	35152.12	
157	NN711988	2004-09-07	63357.13	
161	BR352384	2004-11-14	2434.25	
161	BR478494	2003-11-18	50743.65	
161	KG644125	2005-02-02	12692.19	
161	NI908214	2003-08-05	38675.13	
166	BQ327613	2004-09-16	38785.48	
166	DC979307	2004-07-07	44160.92	

payments 1 x

payments

```

1 SELECT * FROM classicmodels.payments;

```

Result Grid

	customerNumber	checkNumber	paymentDate	amount
145	ED39322	2004-04-26	28211.70	
145	HR182688	2004-12-01	20564.86	
145	JJ246391	2003-02-20	53959.21	
146	FP549817	2004-03-18	40978.53	
146	FU793410	2004-01-16	49614.72	
146	LJ160635	2003-12-10	39712.10	
148	BI507030	2003-04-22	44380.15	
148	DD635282	2004-08-11	2611.84	
148	KM172879	2003-12-26	105743.00	
148	ME497970	2005-03-27	3516.04	
151	BF686658	2003-12-22	58793.53	
151	GB852215	2004-07-26	20314.44	
151	H124	2024-02-07	70.00	
151	IP568906	2003-06-18	58841.35	
151	KI884577	2004-12-14	39964.63	
157	HI618861	2004-11-19	35152.12	
157	NN711988	2004-09-07	63357.13	
161	BR352384	2004-11-14	2434.25	
161	BR478494	2003-11-18	50743.65	

payments 1 x

Exercise 3

Modify the addresses of some customers using the graphical tool, and provide screenshots showing the changes made. Update the records in the customers table as follows:

customerNumber	city	addressLine1	postalCode
141	Girona	43 State St	17001
124	Lleida	2 Walls St	25002
119	Tarragona	46 Marlborough Rd.	43003

- Lines not modified.

Result Grid

	customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
103	Atelier graphique	Schmitt	Carine		40.32.2555	54, rue Royale		Nantes		44000	France	1370	21000.00
112	Signal Gift Stores	King	Jean		7025551838	8489 Strong St.		Las Vegas	NV	83030	USA	1166	71800.00
114	Australian Collectors, Co.	Ferguson	Peter		03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Victoria	3004	Australia	1611	117300.00
119	La Rochelle Gifts	Labrunie	Janine		40.67.8555	67, rue des Cinquante Otages		Nantes		44000	France	1370	118200.00
121	Baane Mini Imports	Bergussen	Jonas		07-98 9555	Erling Skakkes gate 78		Stavem		4110	Norway	1504	81700.00
124	Mini Gifts Distributors Ltd.	Nelson	Susan		4155551450	5677 Strong St.		San Rafael	CA	97562	USA	1165	210500.00
125	Havel & Zbyszek Co.	Piestrzeniewicz	Zbyszek		(26) 642-7555	ul. Filtrowa 68		Warszawa		01-012	Poland		0.00
128	Blauer See Auto, Co.	Keitel	Roland		+49 69 66 90 2555	Lyonerstr. 34		Frankfurt		60528	Germany	1504	59700.00
129	Mini Wheels Co.	Murphy	Julie		6505555787	5557 North Pendale Street		San Francisco	CA	94217	USA	1165	64600.00
131	Land of Toys Inc.	Lee	Kwai		2125557818	897 Long Airport Avenue		NYC	NY	10022	USA	1323	114900.00
141	Euro+ Shopping Channel	Freyre	Diego		(91) 555 94 44	C/ Moralaria, 86		Madrid		28034	Spain	1370	227600.00
144	Volvo Model Replicas, Co	Berglund	Christina		0921-12 3555	Berguvsvagen 8		Luleå		S-958 22	Sweden	1504	53100.00
145	Danish Wholesale Imports	Petersen	Jytte		31 12 3555	Vinbæltet 34		København		1734	Denmark	1401	83400.00
146	Saveley & Henriot, Co.	Saveley	Mary		78.32.3555	2, rue du Commerce		Lyon		69004	France	1337	123900.00
148	Dragon Souvenirs, Ltd.	Natividad	Eric		465 221 7555	Bronz Sok.		NYC		079903	Singapore	1621	103800.00
151	Muscle Machine Inc.	Young	Jeff		2125557413	4092 Furth Circle	Suite 400	Singapore	NY	10022	USA	1286	138500.00
157	Decast Classics Inc.	Leong	Kelvin		2155551555	7586 Pompton St.		Allentown	PA	70267	USA	1216	100600.00
161	Technics Stores Inc.	Hashimoto	Juri		6505556809	9408 Furth Circle		Burlingame	CA	94217	USA	1165	84600.00
166	Handy Gifts Co	Victorino	Wendy		+65 224 1555	106 Linden Road Sandown	2nd Floor	Singapore		069045	Singapore	1612	97900.00

customers 1 x

- Code already modified but not applied.

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale	Nantes	Nantes	44000	France	1370		21000.00
112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.	Las Vegas	NV	83030	USA	1166		71800.00
114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Victoria	3004	Australia	1611	117300.00
119	La Rochelle Gifts	Labruno	Janine	40.67.8555	46 mMariborough Rd.		Tarragona	43003	France	1370		118200.00
121	Baane Mini Imports	Bergulfen	Jonas	07-98 9555	Erling Skakkes gate 78	Stavern		4110	Norway	1504		81700.00
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551480	46 Walls St.		Leids	CA	25002	USA	1165	210500.00
125	Havel & Zbyszek Co	Piestrzeniewicz	Zbyszek	(26) 642-7555	ul. Filbrowa 68		Warszawa	01-012	Poland			0.00
128	Blauer See Auto, Co.	Keitel	Roland	+49 69 66 90 2555	Lyonerstr. 34		Frankfurt	60528	Germany	1504		59700.00
129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street		San Francisco	CA	94217	USA	1165	64600.00
131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue		NYC	NY	10022	USA	1323	114900.00
141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44	43 State St		Grona	17001	Spain	1370		227600.00
144	Volvo Model Replicas, Co	Berglund	Christna	0921-12 3555	Berguvsvägen 8		Luleå	S-958 22	Sweden	1504		53100.00
145	Danish Wholesale Imports	Petersen	Jytte	31 12 3555	Virbæltet 34		København	1734	Denmark	1401		83400.00
146	Saveley & Henriot, Co.	Saveley	Mary	78.32.5555	2, rue du Commerce		Lyon	69004	France	1337		123900.00
148	Dragon Souveniers, Ltd.	Natividad	Eric	465 221 7555	Bronz Sok.		Singapore	079903	Singapore	1621		103800.00
151	Muscle Machine Inc.	Young	Jeff	2125557413	4092 Furth Circle	Suite 400	NYC	NY	10022	USA	1286	138500.00
157	Diecast Classics Inc.	Leong	Kelvin	2155551555	7586 Pompton St.		Allentown	PA	70267	USA	1216	100600.00
161	Technics Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle		Burlingame	CA	94217	USA	1165	84600.00
166	Handy Gifts & Co	Victorino	Wendy	+65 224 1555	106 Linden Road Sandown	2nd Floor	Singapore	069045	Singapore	1612		97900.00

- Applying.

Apply SQL Script to Database

Review SQL Script

Apply SQL Script

Review the SQL Script to be Applied on the Database

```

1 UPDATE `classicmodels`.`customers` SET `addressLine1` = '46 mMariborough Rd.'
2 UPDATE `classicmodels`.`customers` SET `addressLine1` = '46 Walls St', `city` =
3 UPDATE `classicmodels`.`customers` SET `addressLine1` = '43 State St', `city` =
4

```

Back

Apply

Cancel

- Everything has already been implemented.

Limit to 1000 rows

1 • SELECT * FROM classicmodels.customers;

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale		Nantes		44000	France	1370	21000.00
112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.		Las Vegas	NV	83030	USA	1166	71800.00
114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Victoria	3004	Australia	1611	117300.00
119	La Rochelle Gifts	Labrunie	Janine	40.67.8555	46 mMarlborough Rd.		Tarragona		43003	France	1370	118200.00
121	Baane Mini Imports	Bergulfen	Jonas	07-98 9555	Erling Skalkes gate 78		Stavern		4110	Norway	1504	81700.00
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450	46 Walls St		Lleida	CA	25002	USA	1165	210500.00
125	Havel & Zbyszek Co	Piestrzeniewicz	Zbyszek	(26) 642-7555	ul. Filbrowa 68		Warszawa		01-012	Poland		0.00
128	Blauer See Auto, Co.	Keitel	Roland	+49 69 66 90 2555	Lyonerstr. 34		Frankfurt		60528	Germany	1504	59700.00
129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street		San Francisco	CA	94217	USA	1165	64600.00
131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue		NYC	NY	10022	USA	1323	114900.00
141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44	43 State St.		Grona		17001	Spain	1370	227600.00
144	Volvo Model Replicas, Co	Berglund	Christina	0921-12 3555	Berguvsvägen 8		Luleå		S-958 22	Sweden	1504	53100.00
145	Danish Wholesale Imports	Petersen	Jytte	31 12 3555	Vinbæltet 34		Kobenhavn		1734	Denmark	1401	83400.00
146	Saveley & Henriot, Co.	Saveley	Mary	78.32.5555	2, rue du Commerce		Lyon		69004	France	1337	123900.00
148	Dragon Souveniers, Ltd.	Natividad	Eric	+65 221 7555	Bronz Sok.	Bronz Apt. 3...	Singapore		079903	Singapore	1621	103800.00
151	Musde Machine Inc	Young	Jeff	2125557413	4092 Furth Circle	Suite 400	NYC	NY	10022	USA	1286	138500.00
157	Diecast Classics Inc.	Leong	Kevin	2155551555	7586 Pompton St.		Allentown	PA	70267	USA	1216	100600.00
161	Technica Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle		Burlingame	CA	94217	USA	1165	84600.00
166	Handy Gifts & Co	Victorino	Wendy	+65 224 1555	106 Linden Road Sandown	2nd Floor	Singapore		069045	Singapore	1612	97900.00

Exercise 4

Cancel the order made on 2003-09-28, using SQL instructions. Change the status to 'Cancelled', the shippedDate to the current date, and comments to 'Order cancelled due to delay'. Do this using a single SQL statement.

Copy the statement in your submission.

- Statement.

UPDATE orders

SET status = 'Cancelled',

shippedDate = CURDATE(),

comments = 'Order cancelled due to delay'

WHERE orderDate = '2003-09-28';

Exercise 5

Update all product names of type Trains to include the product code in parentheses.

For

example, a product with productCode = S10_1949 and productName = "Vintage Train" will become:

"Vintage Train (code S10_1949)"

Do this with a single SQL statement using the proper MySQL functions.

Resource: MySQL functions - W3Schools

SELECT * FROM products WHERE productLine = 'Trains';

UPDATE products

SET productName = CONCAT(productName, ' (code ', productCode, ')')

WHERE productLine = 'Trains';

Exercise 6

Increase buyPrice and MSRP of all products with quantityInStock > 500 by 0.02%.

Do this with one SQL statement.

UPDATE products

SET buyPrice = buyPrice * 1.0002,

MSRP = MSRP * 1.0002
WHERE quantityInStock > 500;

Exercise 7

Remove all payments made by customers who are represented by employees with the last name 'Patterson'.

Use a single SQL statement and copy it for the task delivery.

```
1. Find employees in Patterson
SELECT employeeNumber
FROM employees
WHERE lastName='Patterson';

2. Customers of those employees
SELECT customerNumber
FROM customers
WHERE salesRepEmployeeNumber IN (...)

3. Delete payments
DELETE FROM payments
WHERE customerNumber IN (...)

4. Final Consultation
DELETE FROM payments
WHERE customerNumber IN (
  SELECT customerNumber
  FROM customers
  WHERE salesRepEmployeeNumber IN (
    SELECT employeeNumber
    FROM employees
    WHERE lastName = 'Patterson'
  )
);
```

Exercise 8

Delete all customers from Lisbon who have not made any payment.

Use one SQL statement only.

```
1. Lisbon Clients
SELECT *
FROM customers
WHERE city='Lisbon';

2. Payment customers
SELECT customerNumber
FROM payments;
```

3. Exclude those customers
NOT IN

4. Final Consultation
DELETE FROM customers
WHERE city = 'Lisbon'
AND customerNumber NOT IN (
 SELECT customerNumber
 FROM payments
);

Exercise 9

Add all customers as new employees, using their contact names as first and last name.

Use customerNumber + 2000 as the new employeeNumber, leave all other fields as 'x0000' extension, 'new@company.com' email, '1' officeCode, 'Sales Rep' jobTitle

Use contactFirstName as firstName, contactLastName as lastName.

Do this with a single SQL statement.

```
INSERT INTO employees (  
employeeNumber,  
lastName,  
firstName,  
extension,  
email,  
officeCode,  
jobTitle  
)  
SELECT  
customerNumber + 2000,  
contactLastName,  
contactFirstName,  
'x0000',  
'new@company.com',  
'1',  
'Sales Rep'  
FROM customers;
```

Exercise 10

Cancel all orders made by customers handled by the customer Elizabeth Lincoln.
Change the status to 'Cancelled', shippedDate to the current date, and comments to 'Order cancelled by management'.
Use a single SQL statement.

1. Find client

```
SELECT customerNumber  
FROM customers  
WHERE contactFirstName='Elizabeth'  
AND contactLastName='Lincoln';
```

2. Cancel orders

```
UPDATE orders  
SET status='Cancelled'  
WHERE customerNumber IN (...)
```

3. Final consultation

```
UPDATE orders  
SET status = 'Cancelled',  
    shippedDate = CURDATE(),  
    comments = 'Order cancelled by management'  
WHERE customerNumber IN (  
    SELECT customerNumber  
    FROM customers  
    WHERE contactFirstName = 'Elizabeth'  
    AND contactLastName = 'Lincoln'  
);
```